

4(a)	$k = F / x$ or $k = \text{gradient}$	C1
	e.g. $k = 4.0 / 0.050$ $k = 80 \text{ N m}^{-1}$	A1
4(b)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$	C1
	$(\Delta)E = (\frac{1}{2} \times 3.2 \times 0.040) - (\frac{1}{2} \times 1.2 \times 0.015) = 0.055 \text{ J}$ or $(\Delta)E = (\frac{1}{2} \times 80 \times 0.040^2) - (\frac{1}{2} \times 80 \times 0.015^2) = 0.055 \text{ J}$ or $(\Delta)E = \frac{1}{2} \times (1.2 + 3.2) \times 0.025 = 0.055 \text{ J}$	A1
4(c)	$(\Delta)E = mg(\Delta)h$	C1
	$= 0.122 \times 9.81 \times (0.120 - 0.095)$ $= 0.030 \text{ J}$	A1
	or	
	$(\Delta)E = W \times (\Delta)h$	(C1)
	$= 1.2 \times 0.025$ $= 0.030 \text{ J}$	(A1)

Question	Answer	Marks
4(d)(i)	$E = 0.055 - 0.030$ $= 0.025 \text{ J}$	A1
4(d)(ii)	$E = \frac{1}{2}mv^2$	C1
	$v = [(2 \times 0.025) / 0.122]^{0.5}$ $= 0.64 \text{ m s}^{-1}$	A1

Question	Answer	Marks
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